Math 540

Homework #4

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§23 Homework Problems

Exercise 4. *Show that if X is an infinite set, it is connected in the finite complement topology.*

Proof. Suppose towards contradiction there exists a separation of X; i.e., we can find $U, V \subseteq X$ open such that $U, V \neq \emptyset$, $U \cap V = \emptyset$, and $U \cup V = X$. Since U, V are open in X and non-empty, it must be the case that their complements are countable. However, note that we can write $X = (U \cap V)^c = U^c \cup V^c$. This is a contradiction, as an infinite set cannot be written as the union of two countable sets. $\square$

Exercise 5. *Show that if X has the discrete topology, then X is totally disconnected. Does the converse hold?*

Proof. Note that for any $a, b \in X$, the set $\{a, b\}$ is not connected as $\{a\}, \{b\}$ form a separation. Let $x \in X$ be arbitrary. Suppose there exists a separation $U, V \in \mathcal{P}(X)$ such that $U, V \neq \emptyset$, $U \cup V = \{x\}$, $\overline{U} \cap V = \emptyset$, and $U \cap \overline{V} = \emptyset$. It must be the case that $U = V = \{x\}$. But $\overline{\{x\}} \cap \{x\} = \{x\} \cap \{x\} = \{x\}$, which is a contradiction. Thus the singletons are the only connected subsets of X.

The converse does not hold. Consider $\mathbb{Q}$ as a subspace of $\mathbb{R}$. The above argument holds identically for elements in $\mathbb{Q}$, however the subspace topology on $\mathbb{Q}$ inherited from $\mathbb{R}$ is not the discrete topology. $\square$

Exercise 9. *Let A be a proper subset of X, and let B be a proper subset of Y. If X and Y are connected, show that $(X \times Y) \setminus (A \times B)$ is connected.*

Proof. Note that for every $x \in X \setminus A$, the set $S_x := \{x\} \times Y$ is connected, and for every $y \in Y \setminus B$, the set $T_y := X \times \{y\}$ is connected. For a fixed $x \in X \setminus A$, note that S_x intersects T_y. Hence $U_x := S_x \cup \left(\bigcup_{y \in Y \setminus B} T_y\right)$ is also connected. Now for two different $x_1, x_2 \in X$, the sets U_{x_1} and U_{x_2} intersect. This means $\bigcup_{x \in X \setminus A} U_x$ is connected. Thus $(X \times Y) \setminus (A \times B)$ is connected, as $\bigcup_{x \in X \setminus A} U_x = (X \times Y) \setminus (A \times B)$. $\square$

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Exercise 1(c). *Show that* $\mathbb{R}^n$ *and* $\mathbb{R}$ *are not homeomorphic if* $n > 1$.

Proof. Suppose there exists a homeomorphism $f : \mathbb{R} \to \mathbb{R}^n$. Let $x \in \mathbb{R}$. Then $f' : \mathbb{R} \setminus \{x\} \to \mathbb{R}^n \setminus \{f(x)\}$ is also a homeomorphism. But $\mathbb{R}^n \setminus \{f(x)\}$ is path-connected, whilst $\mathbb{R} \setminus \{x\}$ is not, which is a contradiction. $\square$

Exercise 3. *Let* $f : X \to X$ *be continuous. Show that if* $X = [0, 1]$, *there is a point* x *such that* $f(x) = x$. *The point* x *is called a* ***fixed point*** *of* f. *What happens if* X *equals* $[0, 1)$ *or* $(0, 1)$?

Proof. Suppose 0 and 1 are not fixed points. Define $g : [0, 1] \to [-1, 1]$ by $g(x) = f(x) - x$. Then $g(0) = f(0)$ and $g(1) = f(1) - 1$. Since 0 and 1 are not fixed points, it must be the case that $g(0) = f(0) > 0$ and $g(1) = f(1) - 1 < 0$. By the Intermediate Value Theorem, there exists an $r \in [0, 1]$ such that $g(r) = 0$. Hence $f(r) - r = 0$; i.e., $f(r) = r$.

Define $f_1 : [0, 1) \to [0, 1)$ by $f_1(x) = \frac{x+1}{2}$. This map is continuous, but does not have a fixed point. Define $f_2 : (0, 1) \to (0, 1)$ by $f_2(x) = x^2$. This map is continuous, but does not have a fixed point. $\square$

Exercise 8.

(a) *Is a product of path-connected spaces necessarily path-connected?*

(b) *If* $A \subset X$ *and* A *is path-connected, is* $\overline{A}$ *necessarily path-connected?*

Proof. (a) Let X, Y be path-connected topological spaces. Let $(a, b), (c, d) \in X \times Y$. Since X is path-connected, there exists a continuous function $f : [0, 1] \to X$ such that $f(0) = a$ and $f(1) = c$. Since Y is path-connected, there exists a continuous function $g : [0, 1] \to Y$ such that $g(0) = b$ and $g(1) = d$. Define $h : [0, 1] \to X \times Y$ by $h(x) = (f(x), g(x))$. This map is continuous (Theorem 18.4), and $h(0) = (a, b)$ and $h(1) = (c, d)$. Thus $X \times Y$ is path-connected.

(b) No. Consider the topologist's sine curve $S := \{(x, \sin(\frac{1}{x})) \mid x \in (0, 1]\}$. This set is clearly path connected, but $\overline{S} = S \cup \{(0, y) \mid y \in [-1, 1]\}$ is not path-connected. $\square$

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Exercise 4. *Let* X *be locally path connected. Show that every connected open set in* X *is path connected.*

Proof. Let $U \subseteq X$ be an open and connected set. Fix $x_0 \in U$ and define:

$$A := \{y \in U \mid \text{ there is a path from } y \text{ to } x_0\}.$$

Give A the subspace topology inherited from U. Note that $A \neq \emptyset$ since $x_0 \in A$.

Let $y \in A$. Then $y \in U$ by inclusion. Since X is locally path connected and since U is open, there exists an open and path connected set V such that $y \in V \subseteq U$. Let $z \in V$. Then there exists a path from z to y. But since there exists a path from y to x_0, there must exist a path from z to x_0; i.e., $z \in A$. Hence $y \in V \subseteq A$, implying that A is open.

Now let $y \in U \setminus A$. Then $y \in U$ by inclusion. Since X is locally path connected and since U is open, there exists an open and path connected set V such that $y \in V \subseteq U$. Suppose towards contradiction we can find some $z \in V \cap A$. Since $z \in V$, there is a path between z and y. Since $z \in A$, there is a path from z to x_0. But this means there is a path from x_0 to y, contradicting the fact that $y \in U \setminus A$. It must be the case that $V \cap A = \emptyset$; i.e., $V \subseteq U \setminus A$. Hence $y \in V \subseteq U \setminus A$, establishing $U \setminus A$ as an open set.

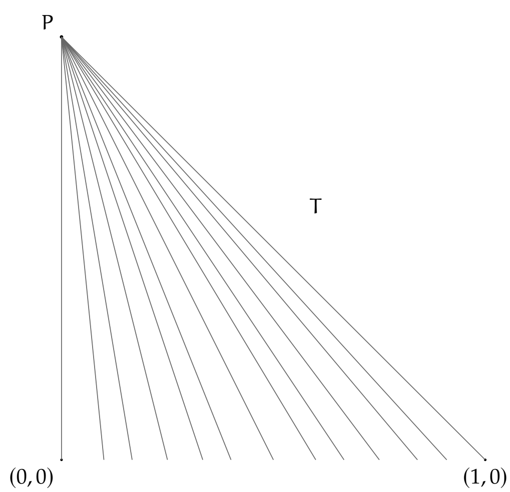
Since A and $U \setminus A$ are open in U, and since U is connected, it must be the case that $A = U$. Thus U is path connected. □

Exercise 5. *Let X denote the rational points of the interval $[0,1] \times \{0\}$ of $\mathbb{R}^2$. Let T denote the union of all line segments joining the point $P = \{0\} \times \{1\}$ to points of X.*

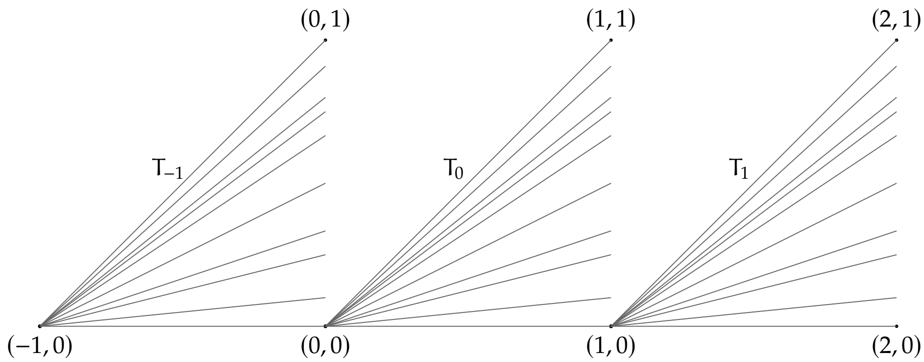
(a) *Show that T is path connected, but is locally connected only at the point p.*

(b) *Find a subset of $\mathbb{R}^2$ that is path connected but is locally connected at none of its points.*

Proof. (a) Let $x, y \in T$. If x and y lie on the same "ray" towards a rational point on $(\mathbb{Q} \cap [0,1]) \times \{0\}$, then surely x and y are path connected. If x and y lie on two different rays, then since x is path connected to p and p is path connected to y, it must be that x is path connected to y. Hence T is path connected. Clearly T is locally connected at p (this can be seen visually). Let $x \in T$ be arbitrary. Let $\epsilon = \frac{d(x,p)}{2}$. Then not only does $B(x, \epsilon)$ contain the ray which x lies on, but by the density of $\mathbb{Q}$ it contains another ray which x does not lie on. As subsets of $B(x, \epsilon)$, these rays are clearly not connected, hence T is locally connected only at the point p.



(b) Consider the set $\mathbb{Z} \times (\mathbb{Q} \cap [0,1])$ as a subset of $\mathbb{R}^2$. Let T_z denote the union of all line segments joining the point $\{z\} \times \{0\}$ to the points in $\{z+1\} \times (\mathbb{Q} \cap [0,1])$. Consider the set $T := \bigcup_{z \in \mathbb{Z}} T_z$.



Note that T is path connected by a similar argument made in (a). However, it is not locally path connected. Indeed, any open ball around $(z, 0)$ not only intersects the rays associated to T_z, but also intersects the rays associated to T_{z-1}. Hence there does not exist a path within any open ball from the point $(z, 0)$ to (z, q), where $q \in \mathbb{Q} \cap [0, 1]$ is nonzero. $\square$